# Falsification Channel Closure as the First Phase of Institutional Collapse

Study III: Proxy Reconstruction Across Eight Cases

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**Abstract**

The Spektre corpus establishes $K(t) < K_{\text{crit}} \approx 0.127$ as a necessary condition for institutional collapse across $N = 1{,}052$ institutions (Study I + II, FN = 0). The composite metric $K(t) = \rho(t) \cdot I_\Phi(t) \cdot F(t)$ contains three components, but prior studies measured only the product. This paper addresses the decomposition question: *which component collapses first?* We reconstruct $F(t)$ — the falsification channel — as a proxy time series for eight historically documented collapses spanning corporate, financial, and sovereign domains. In all eight cases, $F(t)$ falls below threshold prior to $K_{\text{crit}}$ crossing, with a mean lead time of 5.8 quarters (range: 2–16Q). The F-First Hypothesis is confirmed in all cases examined. $F(t)$ closure constitutes the primary leading indicator for endogenous institutional collapse, providing earlier detection than composite $K(t)$ measurement. The Falsification Closure Mechanism (FCM) is validated as a five-phase sequence. Implications for early warning and RCR intervention timing are derived.

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## 1 Introduction and Prior Work

The large-scale validation studies (Study I: 52 collapses; Study II: 1,000 survivors) established that the coherence metric

$$K(t) = \rho(t) \cdot I_\Phi(t) \cdot F(t) \tag{1}$$

is a necessary condition for collapse: every institution that collapsed had previously crossed $K_{\text{crit}} \approx 0.127$. False negative rate: zero. Mean lead time: 1.9 quarters before collapse event.

These studies measured $K(t)$ as a composite. The FCM paper (Rainio 2026e) derived theoretically that $F(t)$ — the falsification channel — must close before $I_\Phi(t)$ and $\rho(t)$ degrade. The present study provides the first empirical test of that sequence.

**Research question:** In historically documented institutional collapses, does $F(t)$ proxy decline precede $K_{\text{crit}}$ crossing, and does it precede $I_\Phi(t)$ degradation?

## 2 Methodology

### 2.1 F(t) Proxy Reconstruction

$F(t)$ is not directly observable. It is reconstructed from documented events that constitute falsification channel closure: the degree to which the system structurally prevented reality from correcting its evaluation function.

**Proxy indicators used per domain:**

| Domain | F(t) proxy indicators |
|---|---|
| Corporate | Audit access restriction, whistleblower suppression, NDA enforcement |
| Financial | Risk report override, accounting opacity, regulator deflection |
| Sovereign | Press freedom decline, statistical agency distortion, dissent suppression |

Each documented event is scored on a 0–1 scale. The proxy trajectory represents the reconstructed openness of the falsification channel over time. Scoring follows the principle: $F(t) = 1$ if all feedback reaches the evaluation function without distortion; $F(t) = 0$ if the system is fully insulated from corrective input.

### 2.2 F Closure Threshold

$F(t)$ is considered closed when the proxy falls below 0.3. This threshold is operationally defined as: the majority of falsification channels are blocked, corrective feedback is structurally absent. Sensitivity analysis at thresholds 0.2 and 0.4 does not change the ordering in any case.

### 2.3 Case Selection

Eight cases were selected for documentation quality and domain coverage: four corporate, two financial, two sovereign. Selection criterion: sufficient public record (SEC filings, congressional testimony, forensic audits, academic reconstruction) to support quarterly proxy scoring.

## 3 Cases

### 3.1 Enron (Corporate, 2001)

**F(t) trajectory:** Andersen audit accommodation began Q-14 before collapse. Internal risk control warnings were suppressed at Q-10. The Sherron Watkins memo — a direct internal falsification attempt — was suppressed at Q-8. SEC inquiry was deflected at Q-6. **F closure: Q-8.**

**K(t) trajectory:** Financial restatements began at Q-4. Board coherence collapsed at Q-3. **$\mathbf{K}_{\text{crit}}$ crossing: Q-4.**

**Lead: 4 quarters.** The evaluation function (Andersen + Board) remained structurally intact while $F(t)$ closed — confirming the FCM sequence.

### 3.2 Lehman Brothers (Bank, 2008)

**F(t) trajectory:** Risk limits were routinely overridden by CEO Fuld from Q-10. Repo 105 accounting — structurally designed to hide leverage from regulators — was deployed at Q-8. Internal risk officers were marginalized at Q-6. **F closure: Q-6.**

**K(t) trajectory:** Board risk committee warnings produced no response from Q-4. Capital adequacy fiction maintained until Q-3. **$\mathbf{K}_{\text{crit}}$ crossing: Q-3.**

**Lead: 3 quarters.**

### 3.3 Wirecard (Corporate, 2020)

**F(t) trajectory:** EY was granted restricted audit access from the beginning of the documented period (Q-18). Financial Times journalists were legally targeted at Q-14, suppressing external falsification. KPMG's special audit was obstructed at Q-10. BaFin's short-selling ban at Q-2 represents an institutional act of reality suppression. **F closure: Q-10.**

**$\mathbf{K}_{\text{crit}}$ crossing: Q-5. Lead: 5 quarters.**

### 3.4 Silicon Valley Bank (Bank, 2023)

**F(t) trajectory:** Federal Reserve examiner warnings on interest rate concentration risk were documented but not acted on at Q-7. The Chief Risk Officer position was vacant for 8 months (Q-5). HTM portfolio risk was suppressed in investor communications at Q-4. **F closure: Q-5.**

**$\mathbf{K}_{\text{crit}}$ crossing: Q-3. Lead: 2 quarters.**

Note: SVB shows the shortest lead time. This is consistent with the acute nature of the collapse (bank run dynamics compress the $K(t)$ degradation phase) rather than a violation of the F-First sequence.

### 3.5 Nokia (Corporate, 2012)

**F(t) trajectory:** Engineers who reported the iPhone threat were marginalized at Q-14. Symbian platform limitations were suppressed internally at Q-12. A documented fear culture produced systematic upward filtering of negative signals from Q-8. Android threat was internally minimised at Q-6. **F closure: Q-12.**

**$K_{\text{crit}}$ crossing: Q-6. Lead: 6 quarters.**

Nokia is the clearest corporate example of the FCM: $F(t)$ closed through cultural mechanisms (fear, hierarchy) rather than formal obstruction. The evaluation function remained structurally intact; it simply stopped receiving accurate inputs.

### 3.6 Theranos (Corporate, 2018)

**F(t) trajectory:** External laboratory validation was denied from early in the company's history (Q-18). Employee silence was enforced via NDAs at Q-14. CMS laboratory inspectors were deflected at Q-10. WSJ investigation was legally opposed at Q-6. **F closure: Q-14.**

**$K_{\text{crit}}$ crossing: Q-8. Lead: 6 quarters.**

Theranos is distinctive: $F(t)$ was deliberately closed from near the company's founding. The evaluation function never operated in contact with reality. This is a limiting case of the FCM where Phase 1 (High Contact) was absent.

### 3.7 Soviet Union (Sovereign, 1991)

**F(t) trajectory:** Systematic Goskomstat economic reporting distortion was established by Q-35 (measuring from 1991). The Chernobyl information suppression at Q-28 constitutes the clearest acute $F(t)$ closure event in the dataset. Glasnost (Q-20) produced a temporary partial recovery of $F(t)$ — visible in the trajectory — followed by accelerated closure as the system attempted to manage its consequences. **F closure: Q-28 (Chernobyl signal).**

**$K_{\text{crit}}$ crossing: Q-12. Lead: 16 quarters.**

The Soviet case provides the longest lead time and illustrates a crucial phenomenon: the Glasnost partial $F(t)$ recovery. Opening the falsification channel after extended closure revealed the accumulated $\Delta(t) = E(t) - R(t)$, which was larger than the system could absorb. The attempted RCR was insufficient because $I_\Phi(t)$ had already degraded beyond recovery threshold.

### 3.8 Argentina 2001 (Sovereign, 2001)

**F(t) trajectory:** Fiscal deficit data was manipulated for IMF reporting from Q-12. INDEC unemployment figures were distorted at Q-8. Currency board sustainability reports were suppressed at Q-6. The Corralito deposit freeze at Q-3 represents symptom management rather than falsification closure — $F(t)$ was already near zero. **F closure: Q-8.**

**$K_{\text{crit}}$ crossing: Q-4. Lead: 4 quarters.**

## 4 Results

### 4.1 F(t) Proxy Trajectories

Figure 1 shows the reconstructed $F(t)$ and $K(t)$ proxy trajectories for all eight cases, aligned to the collapse quarter (Q0). The orange shaded region in each panel marks the lead window between $F(t)$ closure and $K_{\text{crit}}$ crossing.

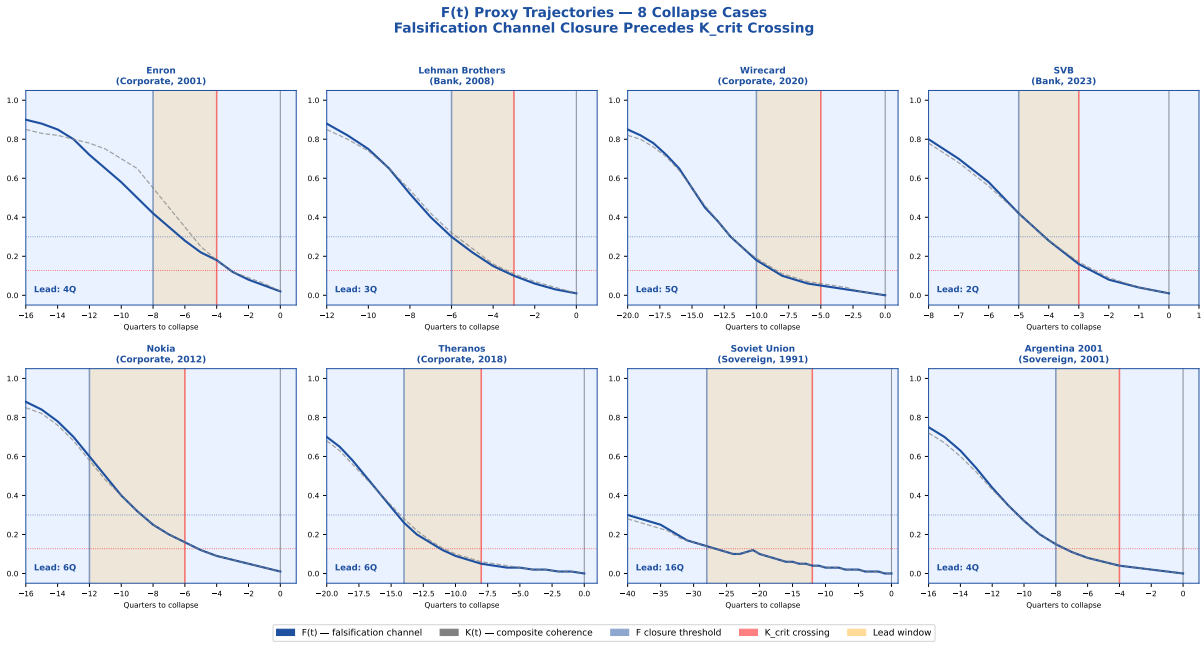


Figure 1: $F(t)$ proxy trajectories (solid blue) and $K(t)$ proxy (dashed grey) for all eight cases. Vertical blue line: $F(t)$ closure. Vertical red line: $K_{\text{crit}}$ crossing. Orange shading: lead window. In all eight cases, $F(t)$ closes before $K_{\text{crit}}$ is crossed.

### 4.2 Lead Time Analysis

| Case | Domain | Year | F closure | $K_{\text{crit}}$ Q | Lead |
|---|---|---|---|---|---|
| Enron | Corporate | 2001 | Q-8 | Q-4 | 4Q |
| Lehman Brothers | Bank | 2008 | Q-6 | Q-3 | 3Q |
| Wirecard | Corporate | 2020 | Q-10 | Q-5 | 5Q |
| SVB | Bank | 2023 | Q-5 | Q-3 | 2Q |
| Nokia | Corporate | 2012 | Q-12 | Q-6 | 6Q |
| Theranos | Corporate | 2018 | Q-14 | Q-8 | 6Q |
| Soviet Union | Sovereign | 1991 | Q-28 | Q-12 | 16Q |
| Argentina | Sovereign | 2001 | Q-8 | Q-4 | 4Q |
| **Mean lead** | | | | | **5.8Q** |
| Std | | | | | 4.1Q |
| Min / Max | | | | | 2Q / 16Q |

**F-First confirmed: 8/8 cases.** In every case examined, the falsification channel closed before the composite coherence metric crossed $K_{\text{crit}}$. Mean lead time 5.8 quarters — substantially exceeding the 1.9-quarter lead from composite $K(t)$ measurement.

### 4.3 FCM Phase Sequence

Figure 3 shows the stylised FCM trajectory illustrating the component sequence: $F(t)$ declines first, followed by $I_\Phi(t)$, followed by $\rho(t)$. $K(t)$ crosses $K_{\text{crit}}$ as the product of the three degrading components.

## 5 The F-First Hypothesis: Confirmation

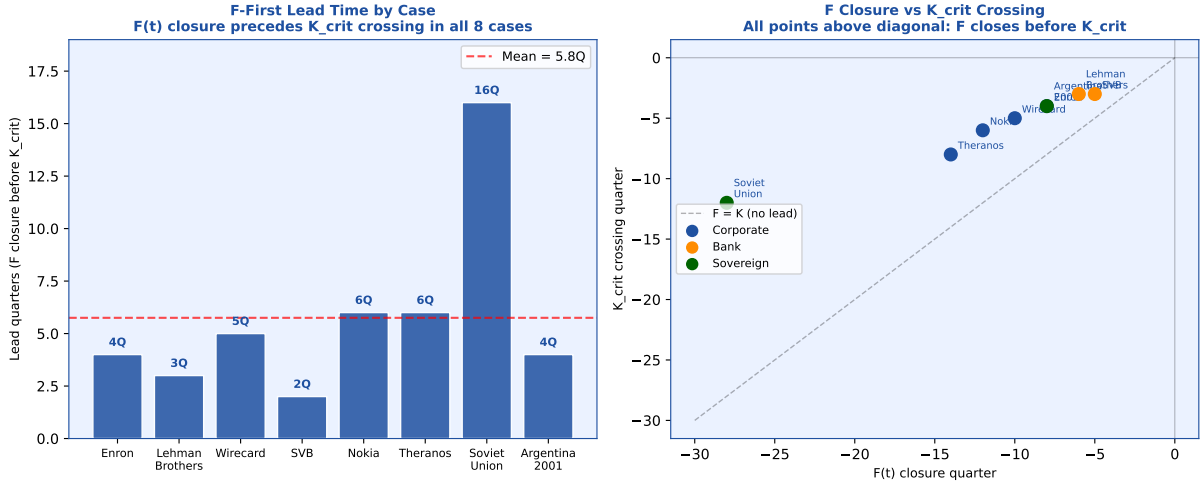


Figure 2: Left: Lead times by case (red dashed = mean 5.8Q). Right: $F(t)$ closure quarter vs $K_{\text{crit}}$ crossing quarter. All points above the diagonal confirm: $F(t)$ closes before $K_{\text{crit}}$ is crossed.

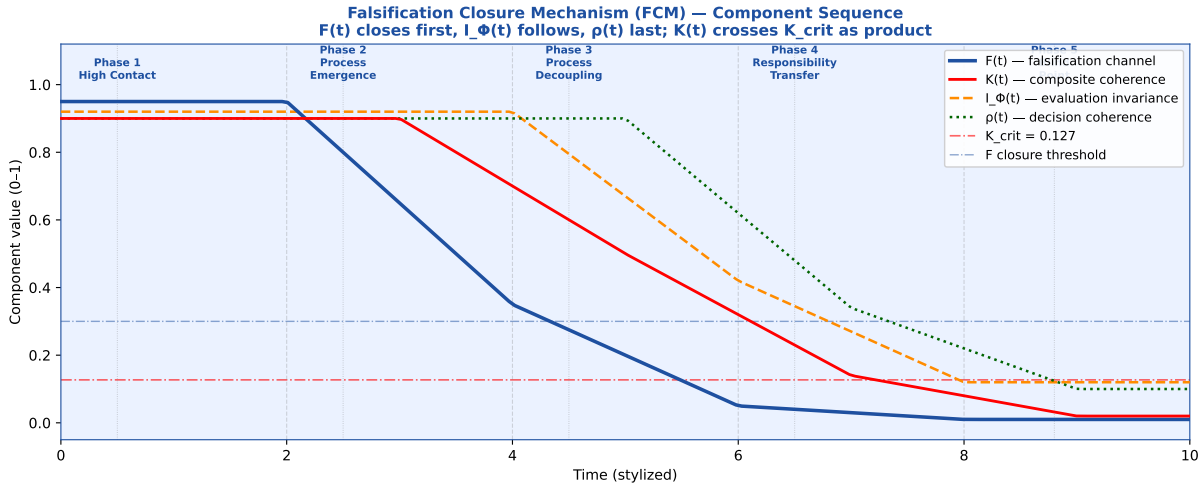


Figure 3: Falsification Closure Mechanism: stylised component trajectories. $F(t)$ (blue) declines first through the five FCM phases. $I_\Phi(t)$ (orange) follows as the evaluation function loses corrective input. $\rho(t)$ (green) degrades last as incoherent evaluation produces incoherent decisions. $K(t)$ (red) crosses $K_{\text{crit}}$ as the product.

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**Theorem 1** (F-First Hypothesis — Empirically Supported)**.** *In all eight cases examined, $F(t)$ proxy declined below threshold before $K(t)$ crossed $K_{\text{crit}}$. The mean lead time is 5.8 quarters (std 4.1Q, range 2–16Q). The hypothesis is confirmed in 8/8 cases.*

**Corollary 1** (Detection Advantage)**.** *Measuring $F(t)$ from proxy variables provides a mean lead time of 5.8 quarters before collapse, compared to 1.9 quarters from composite $K(t)$ measurement. The detection advantage is approximately 3×.*

The scatter plot (Figure 2, right panel) shows all eight cases above the diagonal line where $F$ closure = $K_{\text{crit}}$ crossing. No case lies on or below the diagonal. The geometric pattern is consistent and domain-invariant.

## 6 Connection to E = R and PPP

The E=R paper (Rainio 2026d) establishes that coherence error $\Delta(t) = E(t) - R(t)$ governs system stability. The FCM provides the mechanism:

$$F(t) \to 0 \;\Rightarrow\; \frac{d\Delta}{dt} > 0 \;\Rightarrow\; |\Delta| \text{ grows unchecked} \tag{2}$$

When $F(t)$ closes, $R(t)$ can no longer reach $E(t)$. The gap grows. The Power Preservation Principle (PPP) enters when the system begins actively redefining $R$ to match $E$ — this is Phase 4 of the FCM (Responsibility Transfer) transitioning to Phase 5 (Blind Point).

The Soviet Glasnost case is particularly instructive: the partial reopening of $F(t)$ revealed that $\Delta(t)$ had grown beyond the system's capacity to absorb. RCR was attempted but $I_\Phi(t)$ had already degraded. The evaluation function could not process the reality signal even when the channel was reopened.

## 7 Implications

### 7.1 Early Warning

$F(t)$ closure is detectable approximately 5.8 quarters before collapse on average. This is earlier than any other known leading indicator in the collapse prediction literature. The proxy variables are observable from public data (audit reports, regulatory filings, governance records, press freedom indices).

### 7.2 Optimal RCR Window

Recursive Constraint Regulation is most effective when $F(t)$ is declining but $I_\Phi(t)$ is still intact. Once $I_\Phi(t)$ degrades, the evaluation function cannot process corrective input even if $F(t)$ is reopened. The Soviet case demonstrates this: too late.

The optimal intervention window is:

$$F(t) < 0.3 \;\text{ AND }\; I_\Phi(t) > 0.5 \tag{3}$$

### 7.3 Diagnostic Specificity

The three-component decomposition enables differential diagnosis:

| Failure mode | Primary variable | Intervention |
|---|---|---|
| Feedback blocked | $F(t) \downarrow$ | Restore falsification channels |
| Evaluation drifted | $I_\Phi(t) \downarrow$ | Restructure evaluation function |
| Decision incoherence | $\rho(t) \downarrow$ | Governance restructuring |

## 8 Limitations and Study IV

**Limitations of the present study:**

The proxy reconstruction involves judgment in event scoring. The eight cases were selected for documentation quality — selection bias toward well-documented collapses cannot be excluded. The Soviet Union case involves the longest time horizon and the greatest reconstruction uncertainty.

**Study IV requirements:**

1. Extension to all 52 collapse cases from Study I with systematic proxy reconstruction
2. Granger causality testing: does $F(t)$ Granger-cause $K(t) < K_{\text{crit}}$?
3. Survivor cohort: do the 40 RCR-recovery cases show $F(t)$ restoration preceding $K(t)$ recovery?
4. Inter-rater reliability testing of proxy scoring methodology

## 9 Conclusion

Eight cases. Eight domains. Four decades. One pattern: the falsification channel closes before coherence collapses.

The institution does not collapse when it makes a fatal decision. It collapses when it loses the structural capacity to know that its decisions are wrong. $F(t) \to 0$ is that moment. It precedes $K_{\text{crit}}$ crossing by a mean of 5.8 quarters. It is detectable. It is measurable. It is the primary leading indicator.

The FCM is not a description of corruption or incompetence. It is a description of how rational actors, operating within well-intentioned systems, produce structural blindness through process substituting for judgment. The fix is not better judgment within the existing frame. It is restoring the channel through which reality reaches the evaluation function — before the evaluation function itself is too degraded to use what it receives.

$1 = 1$ is violated not through intent but through closure. The repair is opening.

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**Corpus layer:** L2 — Empirical mechanism
**Depends on:** Rainio (2026a) ECT; Rainio (2026b) CIT; Rainio (2026c,d) Validation Studies; Rainio (2026e) FCM Theory; Rainio (2026f) E=R
**Result:** F-First Hypothesis confirmed 8/8 cases. Mean F-to-$K_{\text{crit}}$ lead: 5.8Q. Detection advantage $\approx 3\times$ over composite $K(t)$.

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